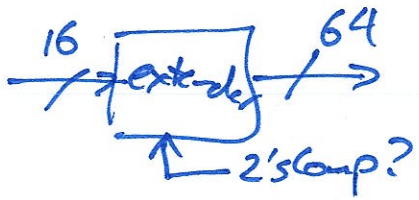
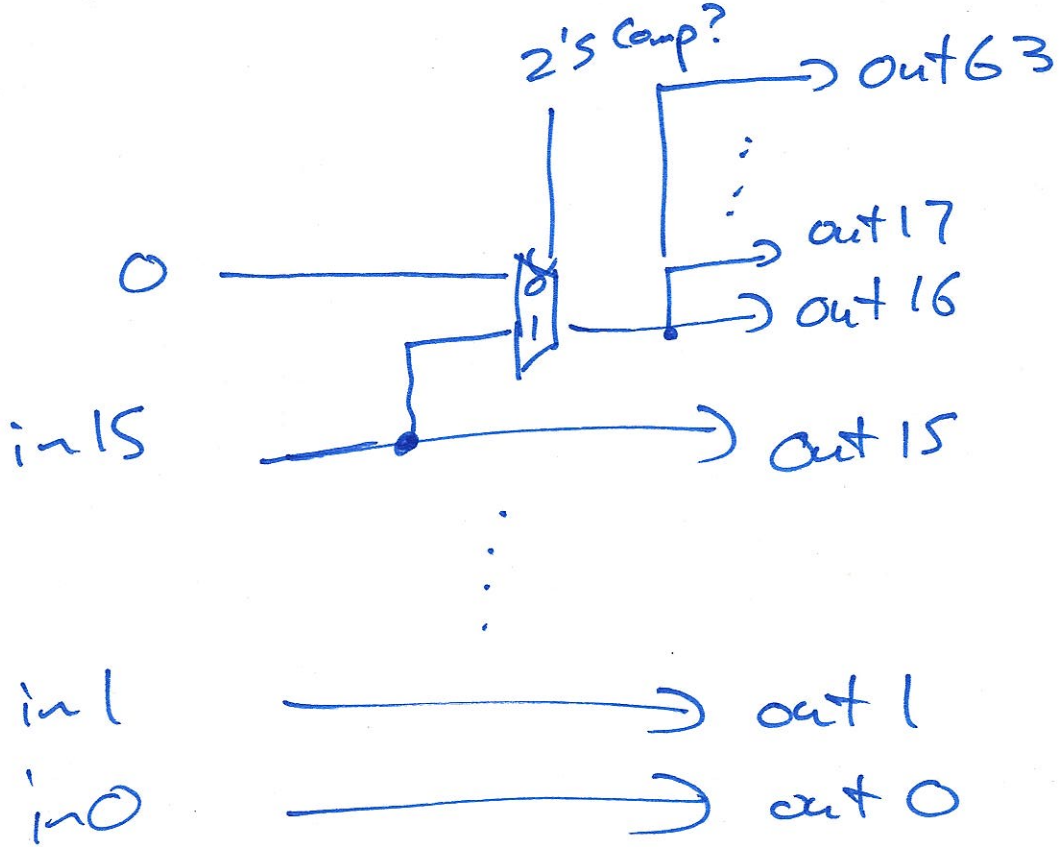
# Review Problem 21




- Immediate vals for some instructions are sign-extended, while others are not. Build a 16bit to 64bit sign-extend unit that can handle both.



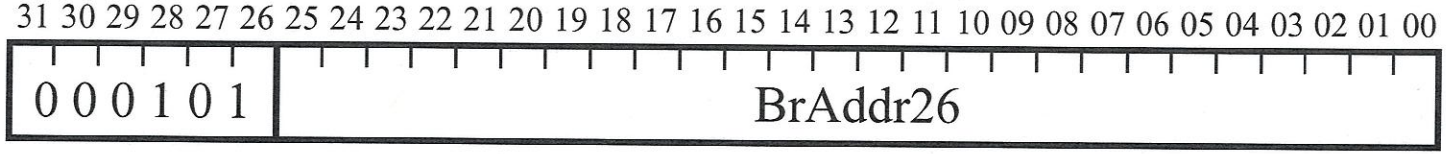
# Branch RTL

Branch Instruction: `B BrAddr26`

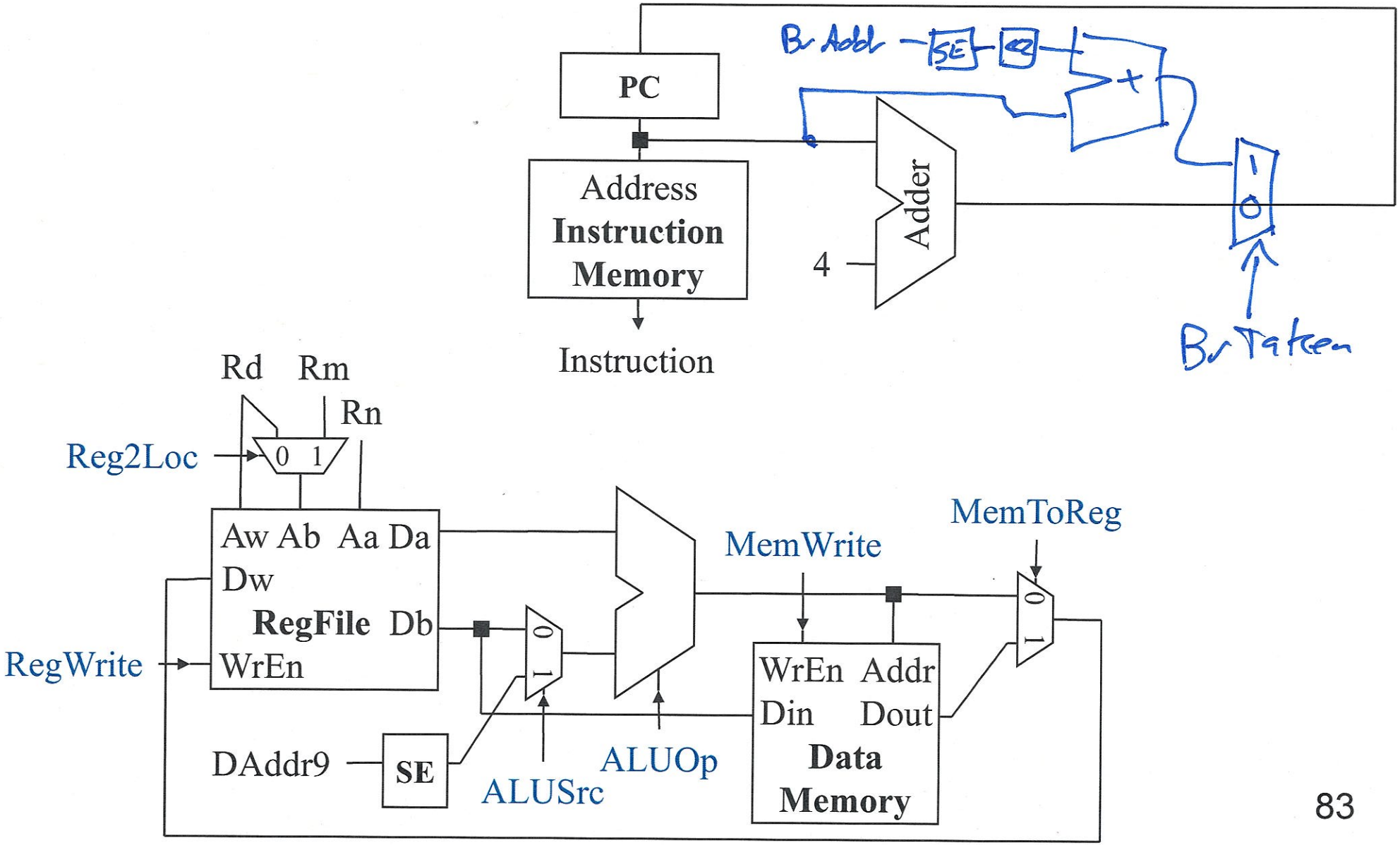
Instruction = MEM[PC];

PC = PC + SE(BrAddr26)<<2

add "00" to the end



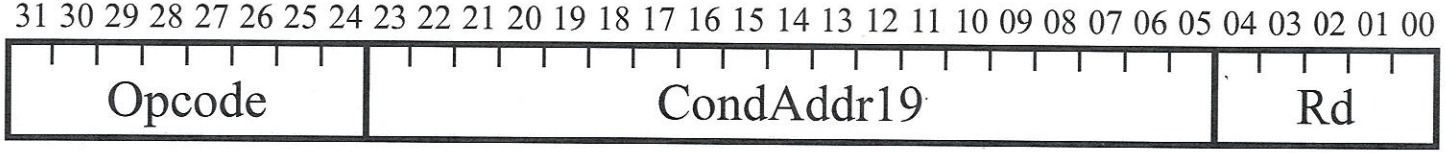
# Datapath + Branch



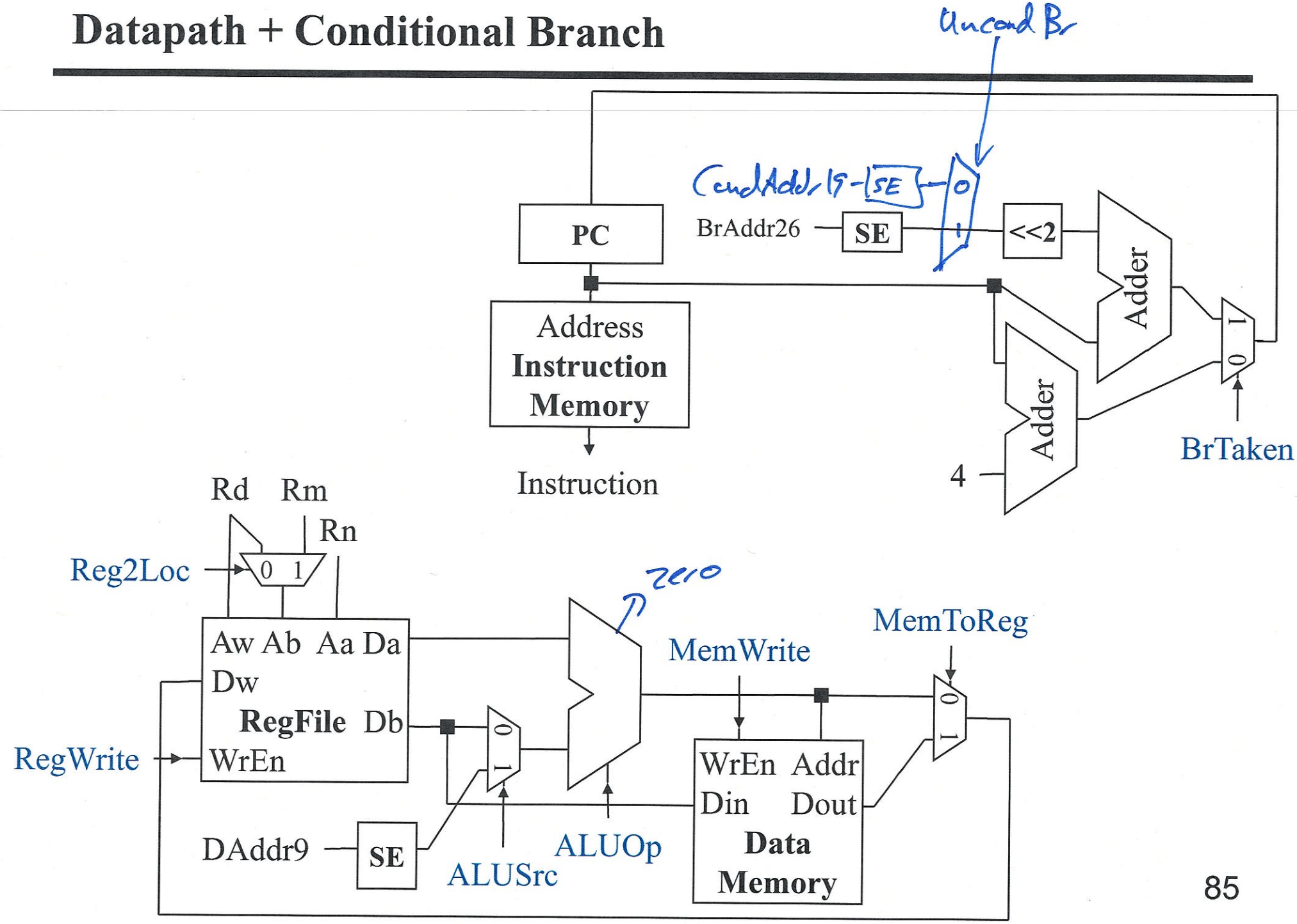
# Conditional Branch RTL

Conditional Branch Instruction: `CBZ Rd, CondAddr19`

```
Instruction = MEM[PC];
Cond = (Reg[Rd] == 0);
if (Cond)
    PC = PC + SE(CondAddr19)<<2;
else
    PC = PC+4;
```



# Datapath + Conditional Branch



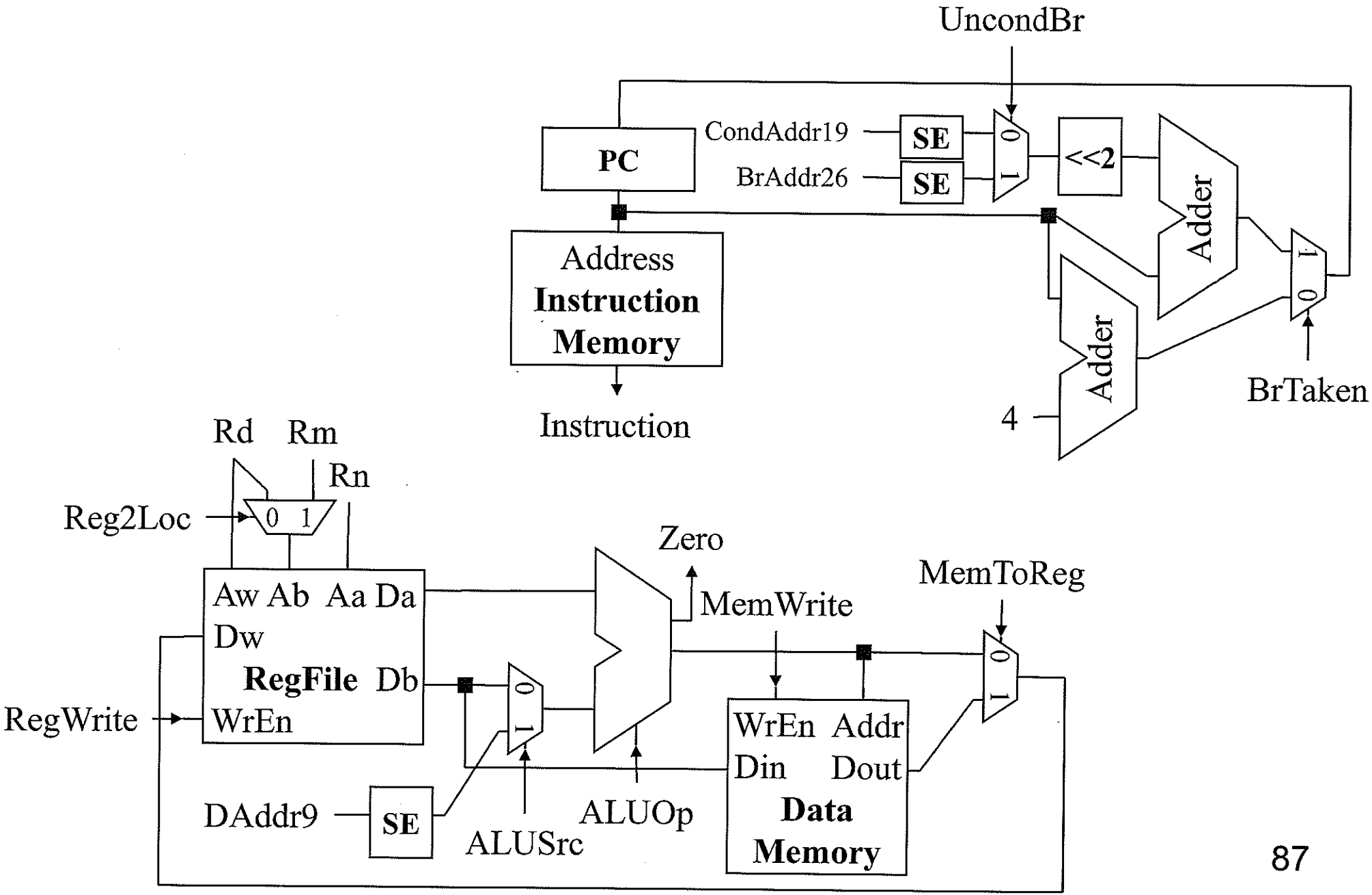
# Control

Identify control points for pieces of datapath

- Instruction Fetch Unit
- ALU
- Memories
- Datapath muxes
- Etc.

Use RTL for determine per-instruction control assignments

# Complete Datapath

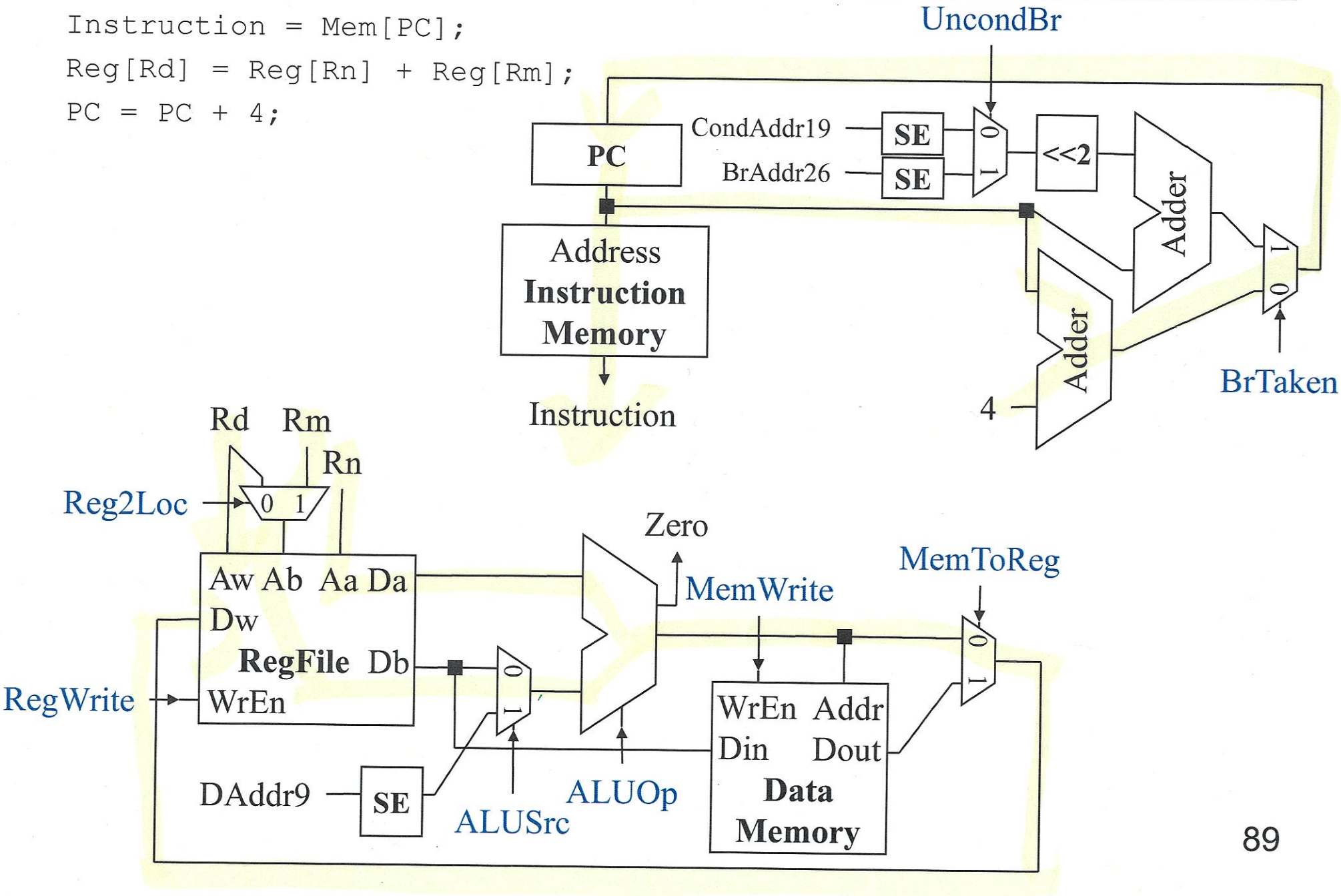


# Control Signals

| Opcode[31:26] Opcode[25:21] | 100010 11000 | 110010 11000 | 111110 00010 | 111110 00000 | 000101 xxxxx | 101101 00xxx |
|---|---|---|---|---|---|---|
| | ADD | SUB | LDUR | STUR | B | CBZ |
| Reg2Loc | 1 | 1 | X | 0 | X | 0 |
| ALU Src | 0 | 0 | 1 | 1 | X | 0 |
| MemTo Reg | 0 | 0 | 1 | X | X | X |
| RegWrite | 1 | 1 | 1 | 0 | 0 | 0 |
| MemWrite | 0 | 0 | 0 | 1 | 0 | 0 |
| Br Taken | 0 | 0 | 0 | 0 | 1 | (zero) |
| UncondBr | X | X | X | X | 1 | 0 |
| ALUOp | + | − | + | + | X | BypassB |

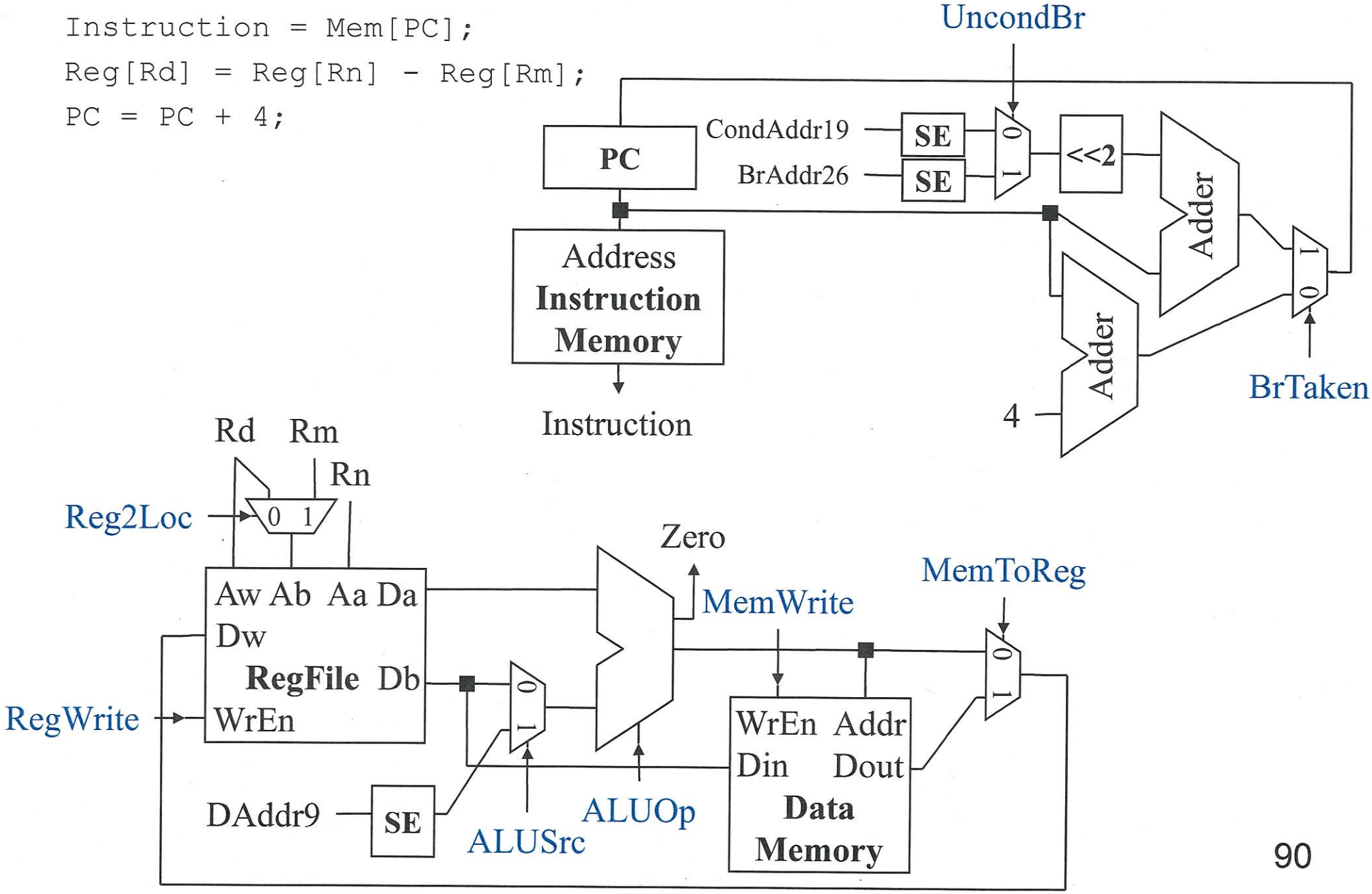
# ADD Control

```
Instruction = Mem[PC];
Reg[Rd] = Reg[Rn] + Reg[Rm];
PC = PC + 4;
```



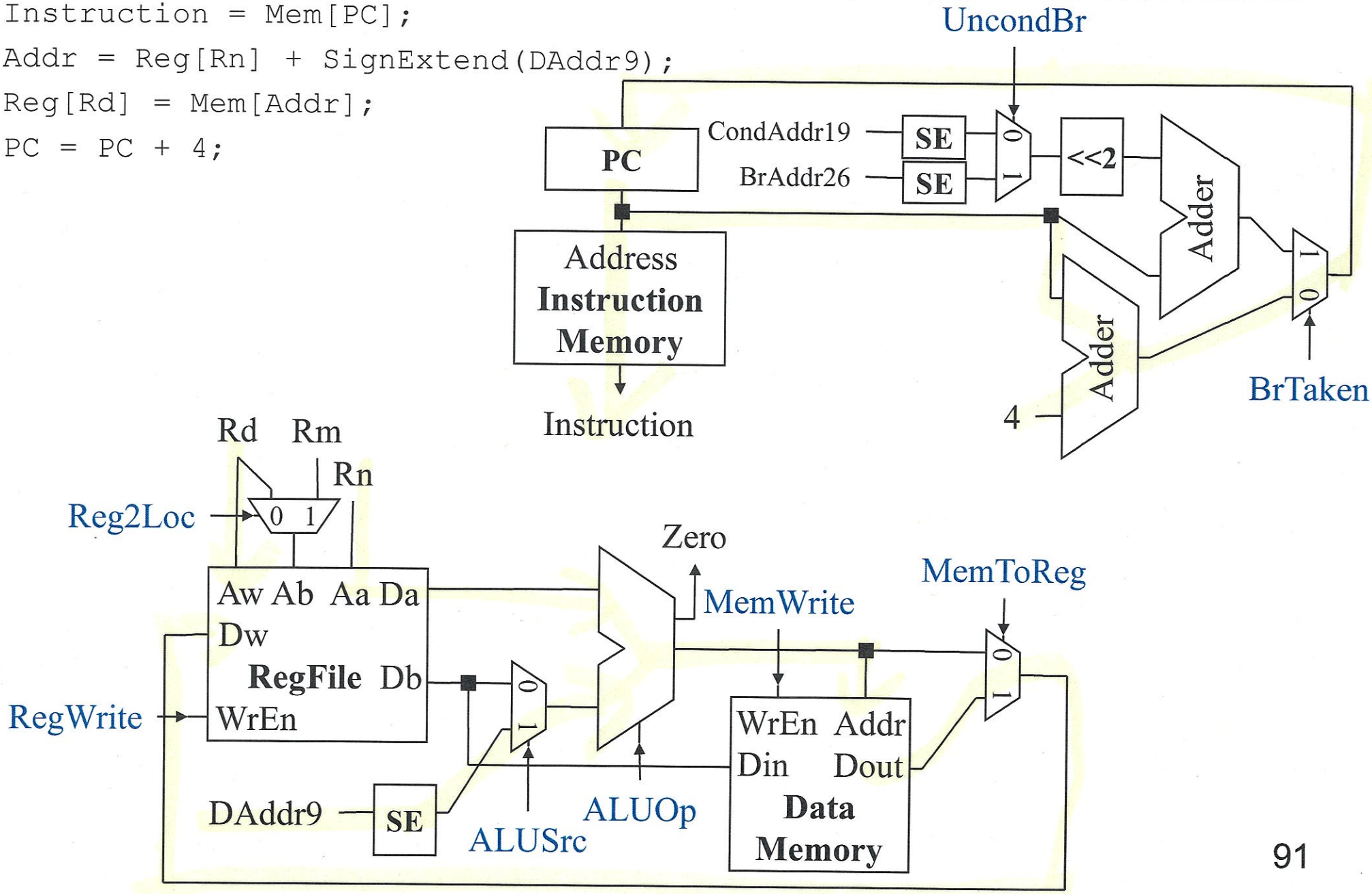
# SUB Control

```
Instruction = Mem[PC];
Reg[Rd] = Reg[Rn] - Reg[Rm];
PC = PC + 4;
```



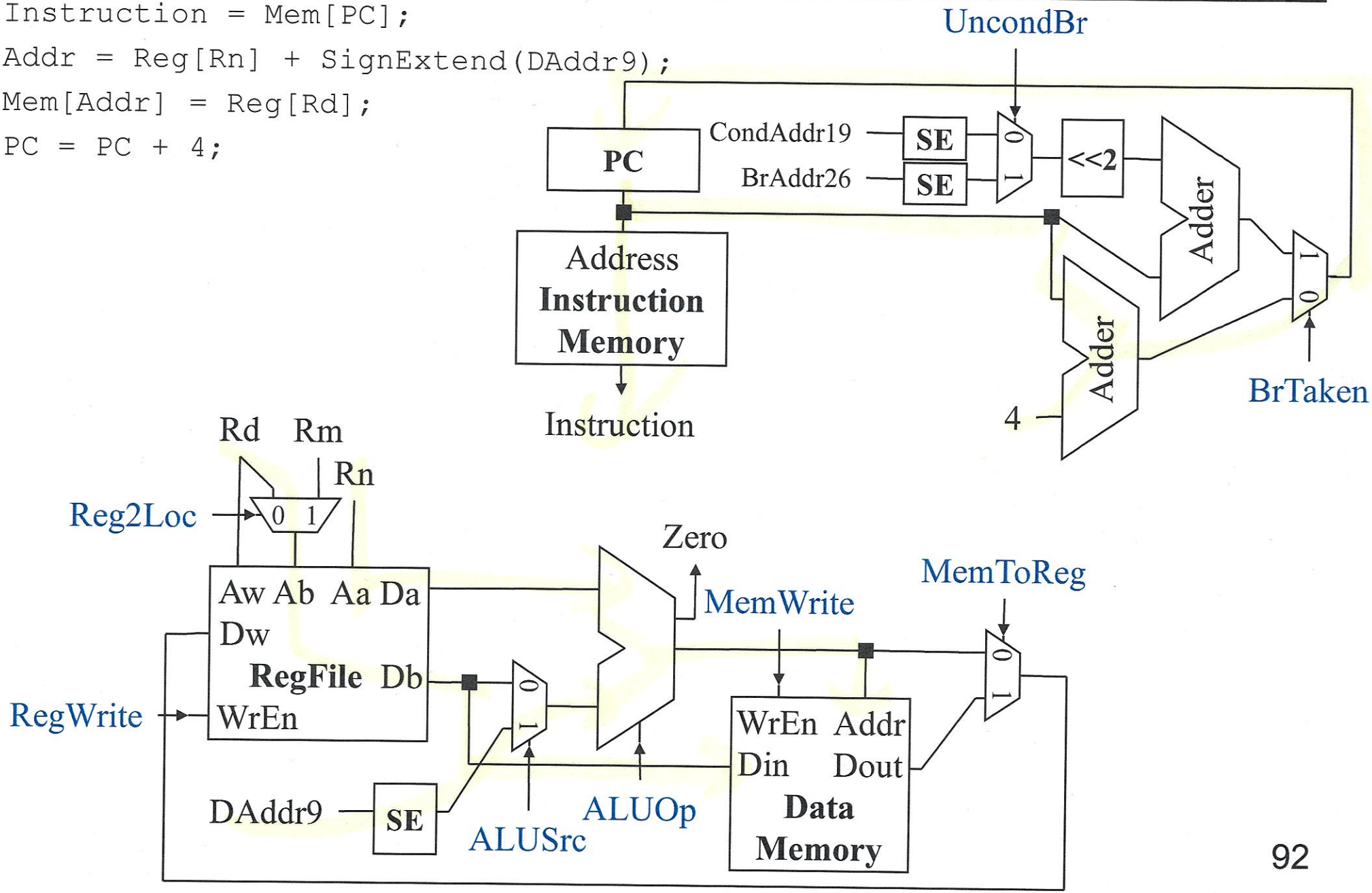
# LDUR Control

```
Instruction = Mem[PC];
Addr = Reg[Rn] + SignExtend(DAddr9);
Reg[Rd] = Mem[Addr];
PC = PC + 4;
```



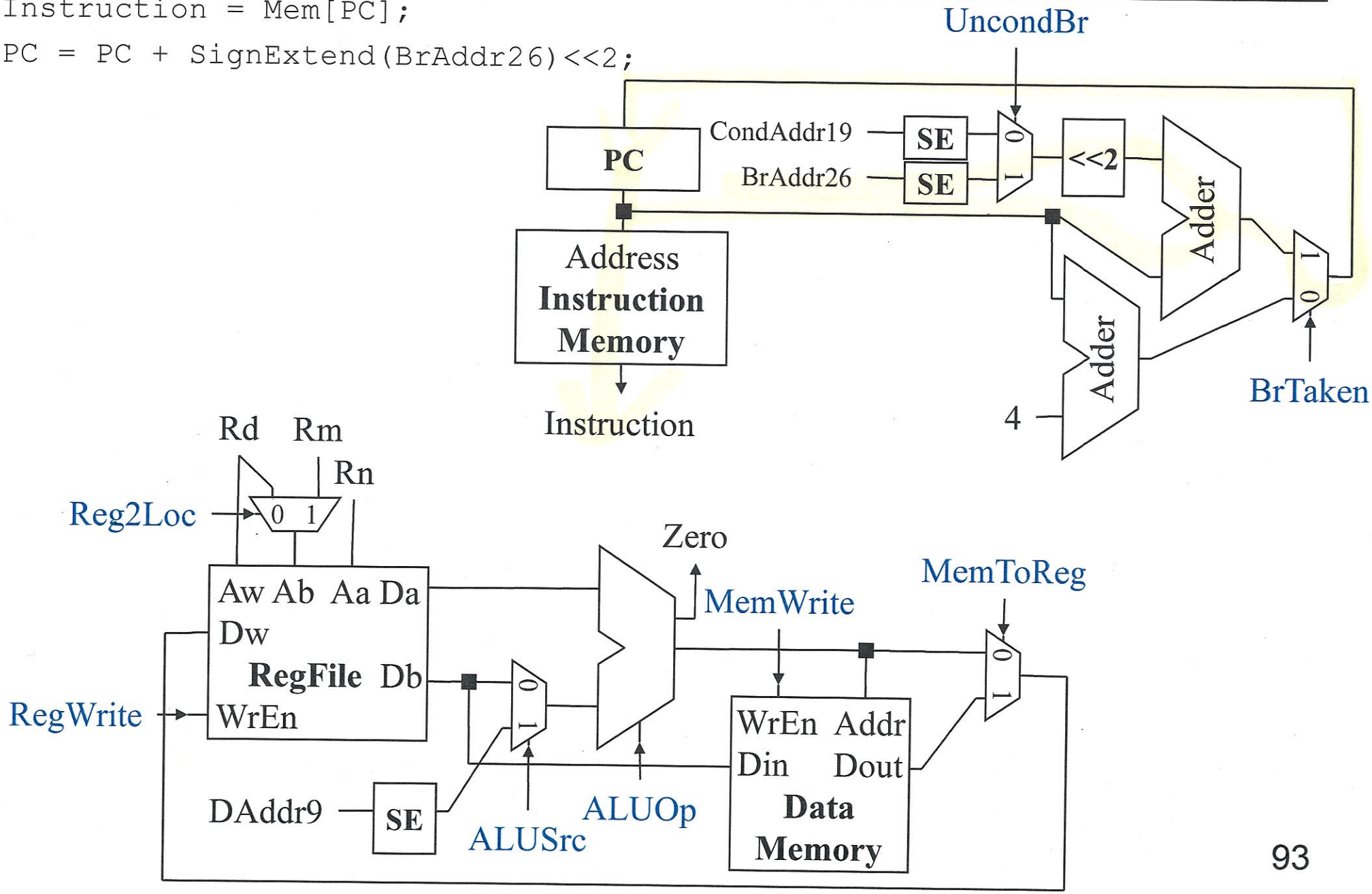
# STUR Control

```
Instruction = Mem[PC];
Addr = Reg[Rn] + SignExtend(DAddr9);
Mem[Addr] = Reg[Rd];
PC = PC + 4;
```



# B Control

```
Instruction = Mem[PC];
PC = PC + SignExtend(BrAddr26)<<2;
```



# CBZ Control

```
Instruction = Mem[PC];
Cond = (Reg[Rd] == 0);
if (Cond)
 PC = PC + SE(CondAddr19)<<2;
else
 PC = PC + 4;
```

